is generally not unique. Hence, only specifying the POVM does not fix the post-measurement state. However, when the post-measurement state is not of interest, it is often convenient to work just with POVMs. In particular, the form of the conditions (2.11)-(2.13) imply that optimisation over arbitrary measurements can be realised using powerful methods of semidefinite programming (a type of convex optimisation). Only the positivity and completeness constraints need to be imposed in practice, because non-negativity implies Hermiticity for operators on a complex Hilbert space.

Projective measurements and observables

An important class of POVMs are those for which the POVM elements are projectors. That is, POVMs with elements $\{\hat{P}_m\}$ such that $\hat{P}_m\hat{P}_n = 0$ for $n \neq m$ and $\hat{P}_n^2 = \hat{P}_n$. These are known as projective measurements and are associated with observables.

An *observable* is a measurable physical property of a system. In quantum mechanics, an observable is any Hermitian operator $\hat{M}$. A Hermitian operator has a spectral decomposition and can be written

$$\hat{M} = \sum_m m\hat{P}_m \tag{2.14}$$

where each m is an eigenvalue of $\hat{M}$ and $\hat{P}_m$ is the orthogonal projector onto the corresponding eigenspace spanned by the eigenvectors with eigenvalue m. The Hermiticity of $\hat{M}$ implies that the eigenvalues are real. An observable thus defines a projective measurement with outcomes given by its eigenvalues (and any projective measurement with real outcomes defines an observable via Eq. (2.14)). For projective measurements, since $\hat{P}_m^\dagger\hat{P}_m = \hat{P}_m^2 = \hat{P}_m$, the POVM elements are themselves measurement operators $\hat{M}_m = \hat{P}_m$ and the (unnormalised) post-measurement state upon obtaining outcome m is $\hat{P}_m|\psi\rangle$. If an eigenvalue m is non-degenerate then $\hat{P}_m$ is the projector onto the corresponding eigenvector, $\hat{P}_m = |m\rangle\langle m|$, and the post-measurement state is then simply $|m\rangle$.

Equivalence of POVMs and observables

From the above, it might seem that POVMs are more general than observables and projective measurements, and POVMs are also sometimes referred to as 'generalised measurements'. In particular, a projective measurement on a d-dimensional system can have at most d outcomes, because projectors corresponding to distinct outcomes must be orthogonal. General POVMs have no such restriction and can have any number of outcomes.

However, any POVM can in fact be realised by letting the system interact unitarily with another, auxiliary system of sufficiently large dimension and then performing a projective measurement on the joint system. In this sense, POVMs and observables are mathematically equivalent. Projective measurements (together with unitary evolution and composition of systems) are sufficient to describe every measurement quantum mechanics can do in principle. In practice, though, it is often much simpler to work with POVMs.